Software Design Description 201

Image Pipeline – Camera to FPGA

Overview

The image pipeline is the primary data path on the DevKitV1. It receives raw grayscale frames from the ESP32-CAM over SPI slave DMA, validates them, queues them, and forwards them to the Tang Nano 9K FPGA over a second SPI master DMA bus. Three modules and two FreeRTOS tasks form a producer-consumer chain with zero-copy handoff through a FreeRTOS queue.

> **Data path:** ESP32-CAM (SPI master) –> DevKitV1 (SPI slave) –> queue –> DevKitV1 (SPI master) –> FPGA (SPI slave)

Table of Contents

Architecture

Wire Protocol

The ESP32-CAM sends frames as a header followed by chunked raw pixel data. The DevKitV1 SPI slave reassembles the stream.

Frame format (CAM –> DevKitV1)

```
Transaction 1: Header (28 bytes)
+-----------+-----------+-----------+-----------+
| magic     | size      | checksum  | timestamp |
| 0xCAFEBEEF| uint32    | XOR       | uint32    |
| 4 bytes   | 4 bytes   | 4 bytes   | 4 bytes   |
+-----------+-----------+-----------+-----------+
| width     | height    | format    | reserved  |
| uint16    | uint16    | uint8     | 7 bytes   |
+-----------+-----------+-----------+-----------+
Total: 28 bytes (IMAGE_HEADER_SIZE)

Transactions 2..N: Data chunks (up to SPI_BUFFER_SIZE each)
```

```
+--------------------------------------------------+
| raw pixel data (up to 4096 bytes per chunk)       |
+--------------------------------------------------+
Repeated until all `size` bytes are transferred.
```

The CAM-side `spi_dma_send_img()` computes an XOR checksum over the full image before sending the header. The header and each data chunk are separate SPI transactions (separate CS assertions).

Header validation

The DevKitV1 validates headers with two checks: 1. `magic == 0xCAFEBEEF` 2. `0 < size <= IMAGE_MAX_SIZE_BYTES` (1 MB)

If either check fails, the header is silently dropped and the state machine stays in `RX_IDLE`.

Checksum

XOR checksum over all `size` bytes of pixel data:

```c
uint32_t checksum = 0;
for (size_t i = 0; i < size; i++)
    checksum ^= data[i];
```

Verified on the DevKitV1 after all data chunks are received. On failure, the entire frame is dropped and the buffer freed.

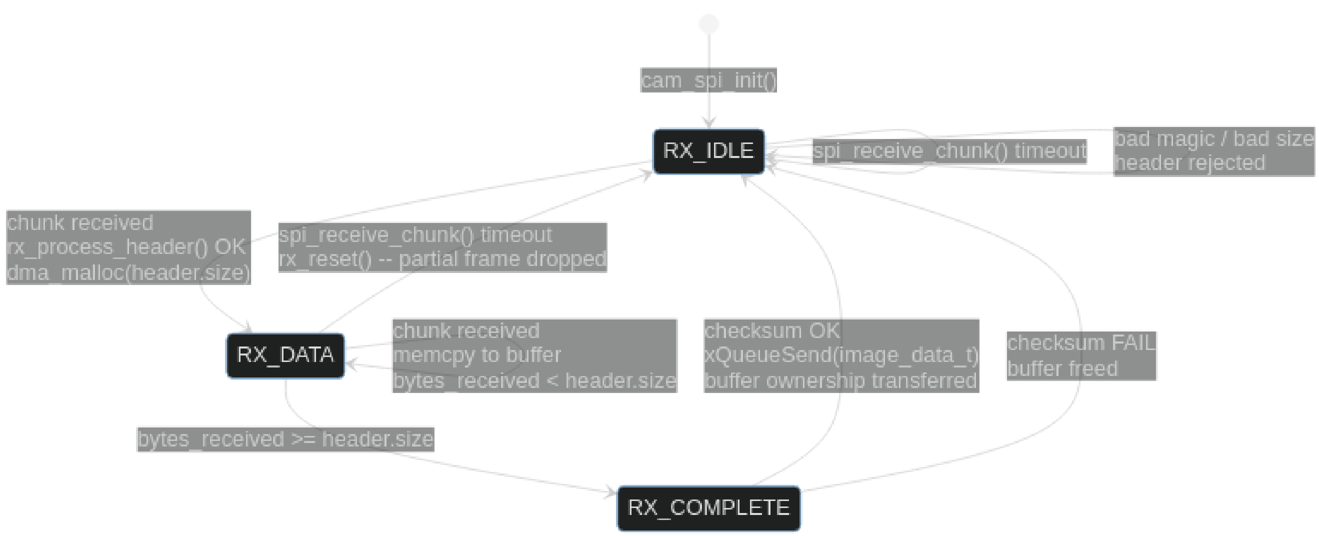
---

cam_spi.c (SPI Slave Receiver)

Responsibilities

- Initialize SPI2_HOST as slave with DMA
- Receive header + data chunks from ESP32-CAM
- Validate header magic and size
- Reassemble chunked data into a contiguous buffer
- Verify XOR checksum
- Enqueue validated `image_data_t` for downstream processing

State machine



SPI slave transaction

Each call to `spi_receive_chunk()` posts one slave transaction and blocks until the master (ESP32-CAM) clocks data in or the timeout expires:

```
spi_slave_transaction_t trans = {
    .length    = SPI_BUFFER_SIZE * 8,   // 4096 bytes max
    .rx_buffer = rx_buf,                // DMA-aligned static buffer
    .tx_buffer = tx_buf,                // MISO (unused, zeroed)
};
spi_slave_transmit(CAM_SPI_HOST, &trans, timeout_ticks);
```

The actual received byte count comes from `trans.trans_len / 8`. This may be less than `SPI_BUFFER_SIZE` if the master sent a short transaction (e.g., the 28-byte header).

Memory management

- `dma_malloc()` allocates a DMA-capable buffer for each incoming frame when the header arrives
- On successful queue insertion, `ctx.buffer` is set to NULL – the queue consumer (`image_processor_task`) owns the allocation
- On queue full or checksum failure, the buffer is freed immediately via `safe_free()`
- The queue holds `image_data_t` structs (buffer pointer + size + header copy), not the pixel data itself

MISO return path (stub)

Line 149 contains a stub for the return path:

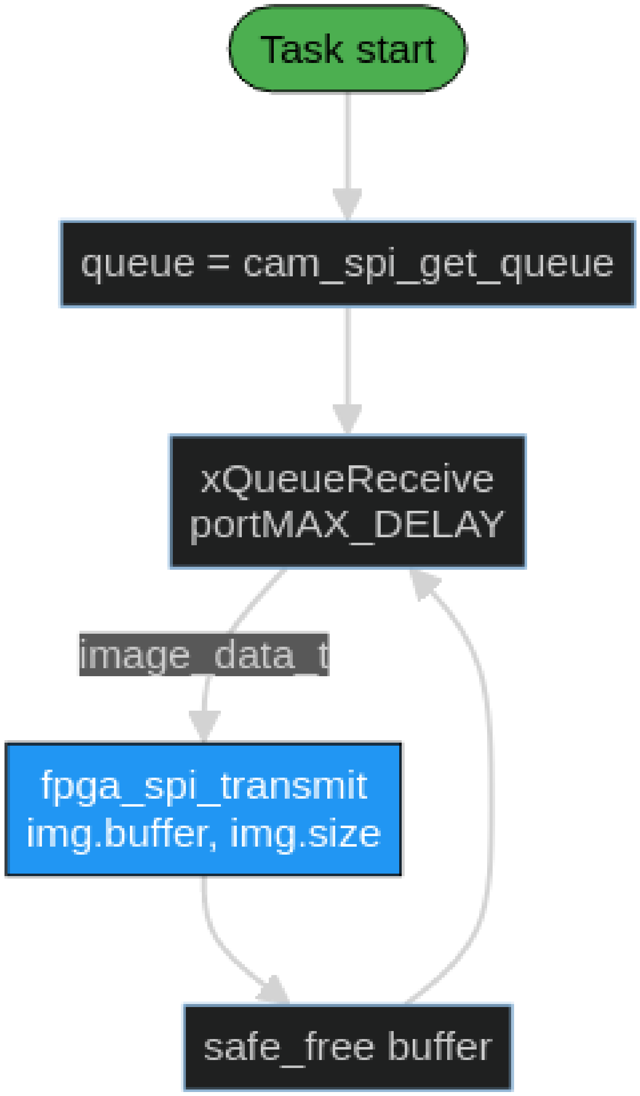
```
cam_spi_send_cmd(uint32_t cmd, uint32_t epoch) {}
```

The ESP32-CAM already reads `tx_buf` during each transaction (full-duplex SPI) and checks for a `0xDEADBEEF` magic in the MISO data. This path is wired but not yet populated – it will carry `CMD_START` / `CMD_STOP` commands from the DevKitV1 to the camera.

---

image_processor.c (Pipeline Bridge)

The simplest module in the pipeline. A single FreeRTOS task that dequeues validated frames and forwards them to the FPGA:



This module is the future insertion point for: - Frame differencing (current - reference) - ML motion detection / hot patch extraction - Selective region forwarding to FPGA

Currently it is a raw passthrough.

---

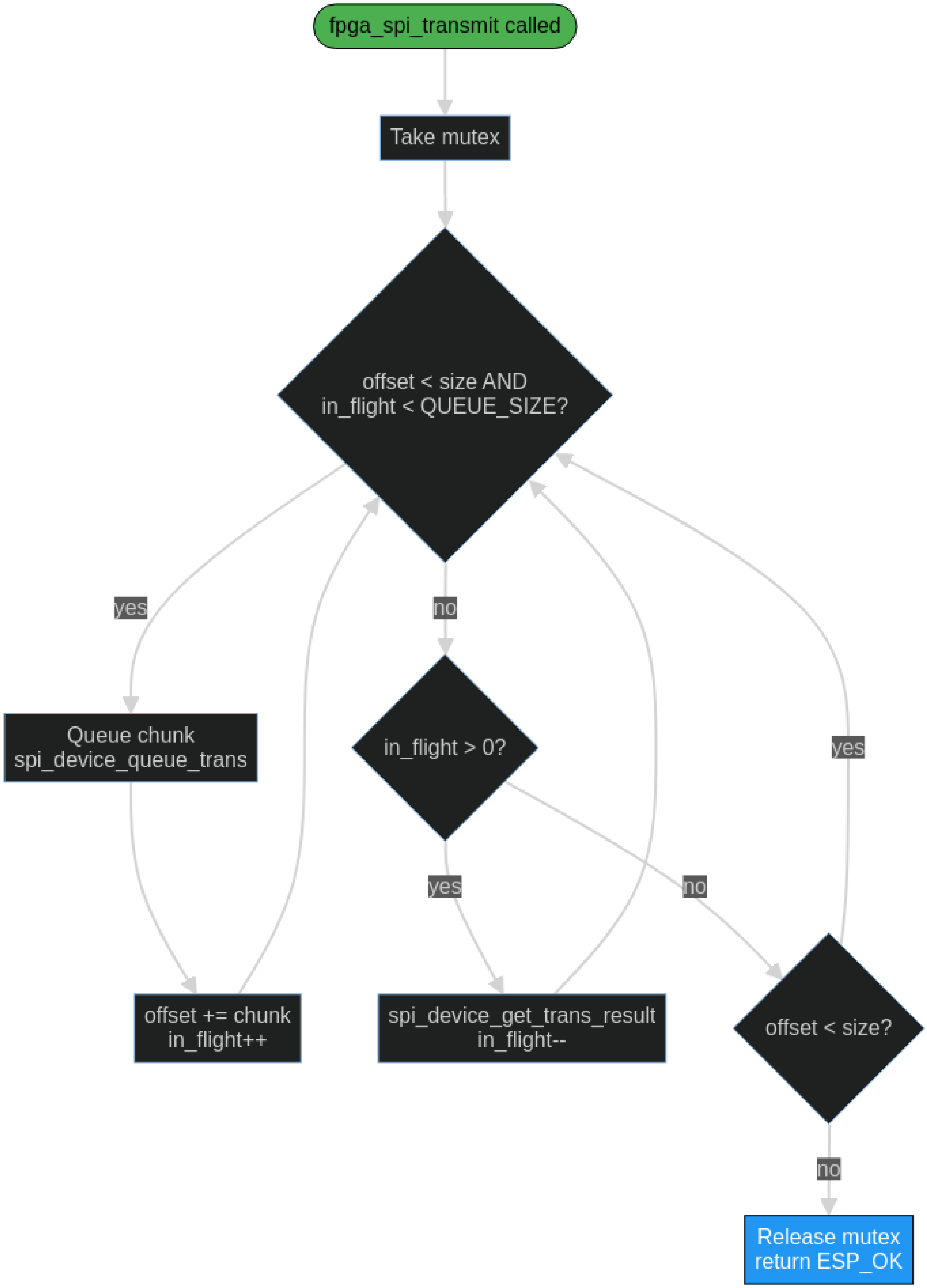
fpga_spi.c (SPI Master Transmitter)

Responsibilities

- Initialize SPI3_HOST as master with DMA
- Transmit image data to FPGA in chunks up to `FPGA_MAX_TRANSFER_SIZE` (32 KB)
- Async queue/get pattern to keep the DMA pipeline fed with no idle gaps
- Mutex-protected access (safe for multiple callers)
- Optional test pattern generation (`TEST_MODE_FPGA_PATTERNS`)

Async queue/get DMA pattern

Unlike the CAM SPI slave (which uses synchronous `spi_slave_transmit`), the FPGA SPI master uses an asynchronous pipeline to maximize throughput. Up to `FPGA_SPI_QUEUE_SIZE` (3) transactions can be in-flight simultaneously:


fpga_spi_transmit called
Take mutex
offset < size AND
in_flight < QUEUE_SIZE?
yes
no
Queue chunk
spi_device_queue_trans
in_flight > 0?
yes
yes
no
offset += chunk
in_flight++
spi_device_get_trans_result
in_flight--
offset < size?
no
Release mutex
return ESP_OK

The pattern works like this: 1. Fill all available queue slots with chunks (up to 3 in-flight) 2. When the queue is full, collect one completed result (freeing a slot) 3. Immediately queue the next chunk into the freed slot 4. Repeat until all data is sent 5. Drain remaining in-flight transactions

This eliminates idle time between DMA transfers – while the SPI hardware is clocking out one chunk, the next chunk is already queued and ready.

Error drain

If any `spi_device_queue_trans` or `spi_device_get_trans_result` fails, the function drains all in-flight transactions before returning. This leaves the SPI queue clean for the next call.

Test patterns (`TEST_MODE_FPGA_PATTERNS`)

When enabled, a test task waits on `g_button_sem` (GPIO0 falling edge ISR) and cycles through four patterns on each press:

Pattern	Description
`PAT_WHITE`	All 0xFF
`PAT_BLACK`	All 0x00
`PAT_GRADIENT`	Horizontal gradient, left=0x00 right=0xFF, 480px period
`PAT_CHECKER`	32x32 pixel checkerboard blocks

Each pattern fills a 32 KB DMA buffer (`TEST_IMAGE_SIZE`) and transmits it as a single CS transaction matching the FPGA BRAM size.

Pin Assignments

CAM SPI (SPI2_HOST – Slave)

Signal	ESP32 GPIO	ESP32-CAM GPIO	Direction
MOSI	GPIO23	GPIO13	CAM –> DevKit
MISO	GPIO19	GPIO12	DevKit –> CAM
SCLK	GPIO18	GPIO14	CAM –> DevKit
CS	GPIO5	GPIO15	CAM –> DevKit

SPI mode 0 (CPOL=0, CPHA=0). CS pin has internal pull-up enabled on the DevKitV1 side.

FPGA SPI (SPI3_HOST – Master)

Signal	ESP32 GPIO	FPGA Pin	Direction
MOSI	GPIO32	Pin 77	DevKit –> FPGA
MISO	GPIO33	Pin 76	FPGA –> DevKit
SCLK	GPIO25	Pin 48	DevKit –> FPGA
CS	GPIO26	Pin 49	DevKit –> FPGA

SPI mode 0, 9 MHz clock. MISO is currently zeroed by the FPGA (`esp_interface.v` line 124) – placeholder for coefficient return path.

Configuration Reference

Macro	Value	Notes
`CAM_SPI_HOST`	SPI2_HOST	Slave bus for camera
`FPGA_SPI_HOST`	SPI3_HOST	Master bus for FPGA
`CAM_SPI_DMA_CHAN`	SPI_DMA_CH_AUTO	IDF auto-selects DMA channel
`FPGA_SPI_DMA_CHAN`	SPI_DMA_CH_AUTO	IDF auto-selects DMA channel
`CAM_SPI_MODE`	0	CPOL=0 CPHA=0
`FPGA_SPI_MODE`	0	CPOL=0 CPHA=0
`FPGA_SPI_CLOCK_MHZ`	9	9 MHz master clock
`CAM_SPI_QUEUE_SIZE`	3	Slave transaction queue depth
`FPGA_SPI_QUEUE_SIZE`	3	Master async pipeline depth
`SPI_BUFFER_SIZE`	4096	Per-chunk receive buffer (cam slave)
`FPGA_MAX_TRANSFER_SIZE`	32768	Max single master transaction (BRAM size)
`CAM_TRANSFER_TIMEOUT_MS`	5000	Slave receive timeout
`IMAGE_MAX_SIZE_BYTES`	1048576	Max accepted frame size (1 MB)
`IMAGE_QUEUE_LENGTH`	3	Queue depth between cam_rx and img_proc
`IMAGE_HEADER_MAGIC`	0xCAFEBEEF	Frame header magic word
`IMAGE_HEADER_SIZE`	28	Header struct size (bytes)

Source Files

File	Lines	Purpose
`src/interprocessor/cam_spi.c`	230	SPI slave receiver, RX state machine, checksum, queue
`src/utilities/image_processor.c`	47	Queue consumer, raw passthrough to FPGA
`src/interprocessor/fpga_spi.c`	260	SPI master transmitter, async DMA, test patterns
`inc/cam_spi.h`	43	Public interface: init, deinit, get_queue, receive_task
`inc/fpga_spi.h`	47	Public interface: init, deinit, transmit, test_task
`inc/image_processor.h`	17	Public interface: init, task

References

- SDD200 – DevKitV1 Overview
- SDD100 – ESP32-CAM – transmit side of the SPI link
- SDD400 – FPGA Subsystem – `esp_interface.v` receive side
- ESP-IDF SPI Slave API
- ESP-IDF SPI Master API